

Factors Influencing the Median Home Value Across U.S. Metropolitan Areas

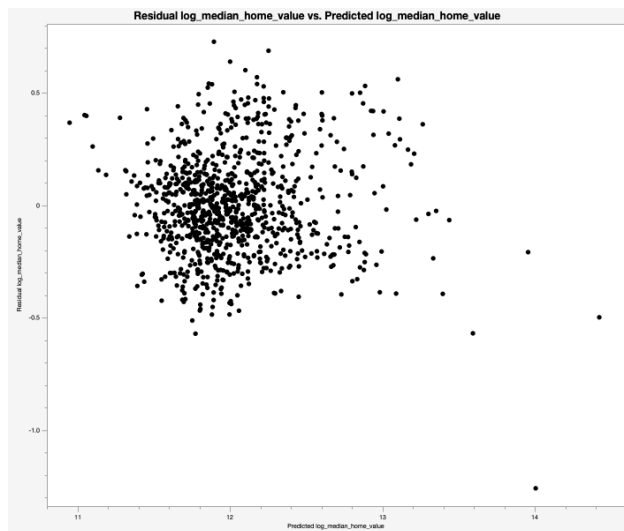
ILRST 2130 Project Component 4

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In this component, regression diagnostics are conducted to evaluate the validity of the assumptions underlying the multiple linear regression model fitted in Component 3. The analysis focuses on the log-transformed response variable, log median home value, and the same set of economic, demographic, and housing market predictors previously included in the model. The purpose of this phase is to assess whether the assumptions of linearity, constant variance, and normality of residuals are reasonably satisfied, and to determine whether any observations exert disproportionate influence on the fitted regression equation.

Specifically, residual plots are examined to evaluate linearity and homoscedasticity, while studentized residuals, leverage values, Cook's Distance, and DFFITS are used to identify potential outliers and influential observations. In addition, histograms and normal quantile plots of the residuals are analyzed to assess the normality assumption. These diagnostic tools will provide a comprehensive assessment of model adequacy and help determine whether the regression results can be interpreted reliably

Ordinary residuals against Fitted Values



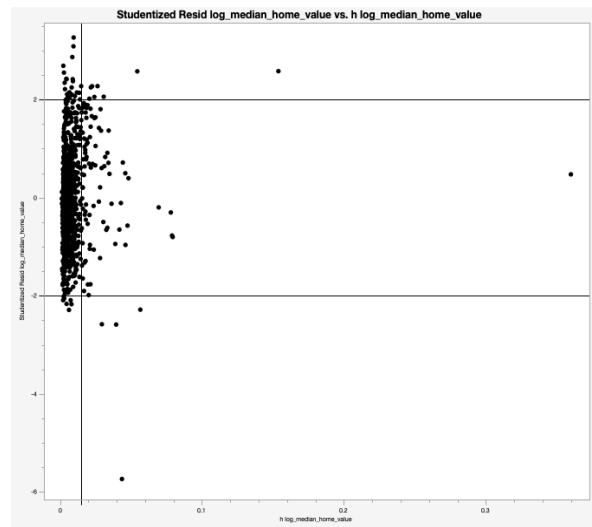
The residuals versus fitted values plot for the log-transformed model shows residuals that are approximately randomly scattered around zero with no clear curvature. This indicates that the linearity assumption appears reasonable. In contrast to the original model, there is no strong funnel-shaped pattern, suggesting that the variance of the residuals is more constant across fitted values. Although a few observations exhibit relatively large residuals at higher predicted values, these points do not form a systematic pattern. Overall, the residual plot suggests that the log-transformed model better satisfies the assumptions of linear regression.

Residuals vs Leverage Values

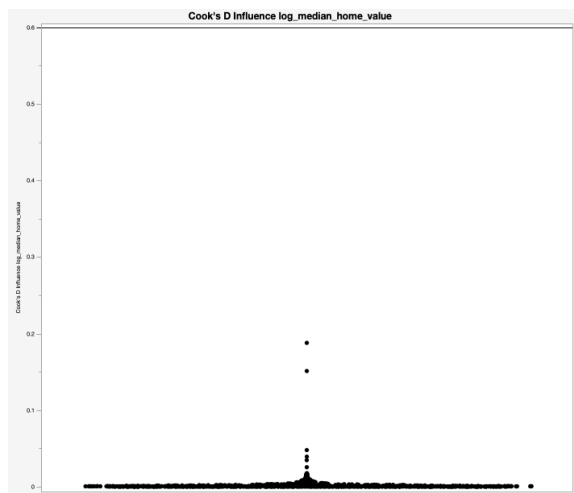
The studentized residual versus leverage plot was examined to identify outliers and high leverage points. Using the rule of thumb $|r_i| > 2$, several observations exceed this threshold, indicating the presence of moderate outliers, including one particularly extreme negative residual near -5.8 . However, given the large sample size ($n=939$), this is not unexpected, as approximately 5% of observations would naturally

fall beyond ± 2 under the assumption of normality. The leverage cutoff was determined using $h > 2(p/n)$. With 7 predictors and $n=939$, leverage is considered large when $h > 0.015$. The majority of observations fall well below this threshold. Although one observation exhibits substantially higher leverage (approximately 0.33), its studentized residual is relatively small in magnitude. This suggests that, despite its unusual predictor values, it does not strongly distort the fitted regression line.

Additionally, most observations that exceed ± 2 have low leverage values and therefore are unlikely to exert substantial influence on the fitted regression model. Only a small number of observations exhibit both large residual magnitude and elevated leverage, suggesting that overall influence is limited.



Cook's Distance plot

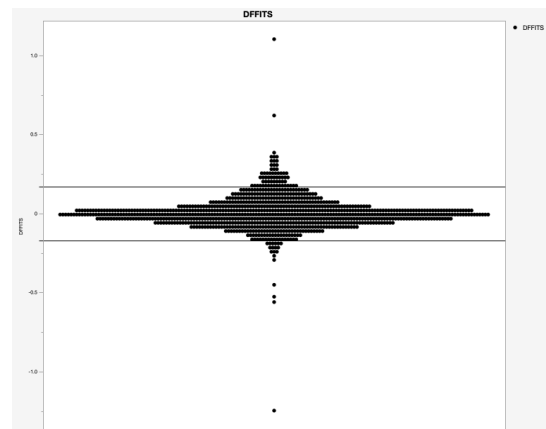


Cook's Distance was examined to assess the overall influence of individual observations on the fitted regression model. Using the rule of thumb based on the 20th percentile of the F distribution with 7 and 932 degrees of freedom, the Cook's Distance cutoff is approximately 0.60. The vast majority of Cook's Distance values are very close to zero, indicating minimal influence for most observations. Although a small number of points exhibit moderately larger Cook's Distance values (around 0.15–0.18), none exceed the 0.60 threshold. This suggests that while a few observations have some measurable impact on the fitted values, there are no highly influential cases that

substantially distort the regression results. Overall, the Cook's Distance diagnostics support the stability of the fitted model.

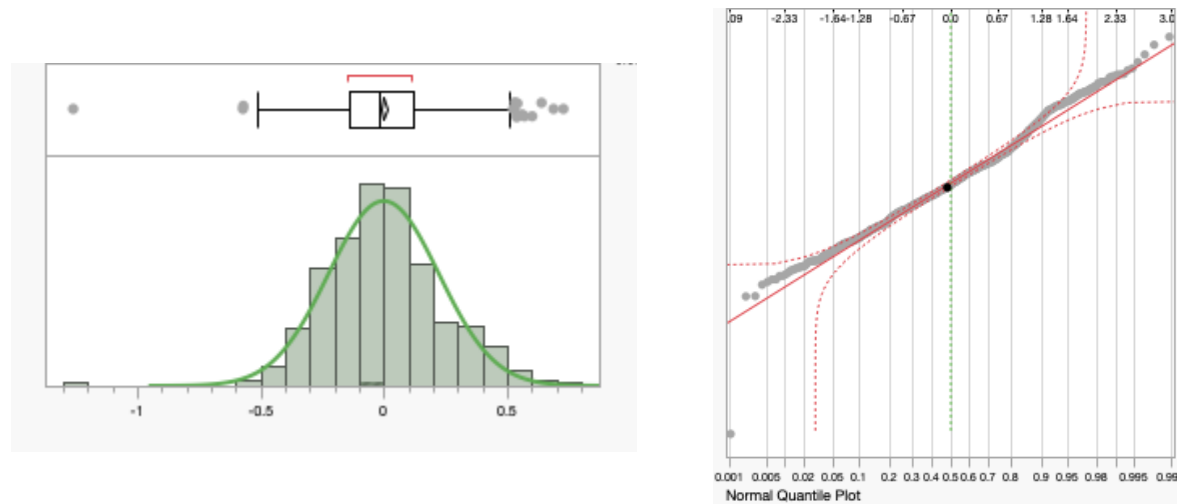
DFFITS plot

The DFFITS diagnostic was examined to evaluate the influence of individual observations on their fitted values. For large datasets ($n \geq 100$), the rule of thumb $|DFFITS| > 2p/n$ to identify potentially influential cases. With $p=7$ predictors and $n=939$, the cutoff is approximately ± 0.17 . The majority of DFFITS values cluster tightly around zero, indicating that most observations have minimal impact on their predicted



values. However, a small number of observations exceed the ± 0.17 threshold, including one notably large negative value and one large positive value. These points suggest that removing these observations would meaningfully change their fitted values. Nevertheless, only a few observations exceed the reference cutoff (given that $n=939$), indicating that overall model influence is limited and the regression results are not driven by a substantial number of highly influential cases.

Histogram of Ordinary Residuals & Normal Quantile Plot



The histogram of the ordinary residuals indicates that the residuals are approximately symmetric and centered around zero, with a distribution that is reasonably bell-shaped. Although a few extreme values appear in both tails, the overall shape aligns closely with the overlaid normal curve. The normal quantile (Q-Q) plot further supports this assessment, as most residuals lie near the reference line, with only minor deviations observed in the extreme tails. These small departures suggest the presence of a few outlying observations rather than a systematic violation of normality. Given the large sample size ($n=939$), such mild tail deviations are not unexpected. Overall, the residual diagnostics do not indicate substantial departures from the normality assumption.